END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

B. Tech (CSAI/ECE AI), Semester: II[nd]

(AUGUST, 2022)

Subject Code: BAS 106	Subject: Environmental Science
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			(2.5*8=20)
	(a)	What is octane number and discuss its relation with the chemical structure of the fuel?	
	(b)	Explain the terms TD50, ED50 and LD50 with the help of dose-response curve.	
	(c)	Discuss the environmental parameters considered under Environment Impact Assessment (EIA).	
	(d)	What is greenhouse effect and how it is related with global warming?	
	(e)	Briefly discuss the synthesis and importance of biodiesel.	
	(f)	Explain the negative consequences of modern agriculture.	
	(g)	What are promoters or photo sensitizers? Discuss their role in biodegradation of polymers.	
	(h)	Discuss carbon dioxide sequestration along with the concept of carbon credits.	
		UNIT-I	
Q2		What are the different types of minerals? Describe the methods adopted for the extraction of minerals and how mineral extraction effects the environment?	(10)
Q3		Why forests are economically and ecologically important? Also explain deforestation along with its consequences.	(10)
		UNIT-II	
Q4		Discuss any five principles of Green chemistry with suitable examples.	(10)
Q5		Explain the various approaches involved in waste water treatment.	(10)
		UNIT-III	
Q6		List out the various alternate energy sources. Explain the energy harnessing from - (a) solar energy (b) Hydro energy	(10)
Q7		A sample of coal was found to have the following percentage composition: C = 75%; H_2 = 5.2%; O_2 = 12.1%; N_2 = 3.2% and ash = 4.5%. (a) Calculate the minimum air required for the complete combustion of 1 Kg of coal. (b) Also calculate the higher and lower calorific value of coal sample using Dulong's Formula. (Gross calorific value in Kcal/Kg: C = 8080; H = 34500; S = 2240)	(10)
		UNIT-IV	
Q8	(a)	What are Eco-friendly polymers? Give example of any two types of bioplastictics.	(5,5)
	(b)	Discuss photo-biodegradation and hydro-biodegradation of polymers.	
Q9	(a)	Discuss the various routes for the entry of toxic materials in the body.	(5,5)
	(b)	Explain the biochemical effect of lead poisoning on human body.	